PrincePDF Documentation

A Fast, Modern, and Minimalist PDF Viewer for Windows 11

Overview

PrincePDF is engineered for speed, clean aesthetics, and seamless Windows 11 integration. Built with PyMuPDF and PyQt6, it delivers instant page rendering, buttery-smooth scrolling, and pixel-perfect typography on High-DPI displays.

Key Features:
- Lightning-fast document loading and responsive page caching
- Single Page, Continuous Scroll, and Side-by-Side (Spread) view modes
- Accurate text selection and copy to clipboard (Ctrl+C)
- In-document search with match highlighting and next/prev navigation
- Dark Mode UI and Night Reading Mode (color inversion)
- Document Outline / Table of Contents navigation
- Page Thumbnails sidebar for rapid jumping
- Windows 11 Default PDF Viewer integration

Keyboard Shortcuts & Controls

Navigation:
- Next Page: Page Down / Down Arrow / Right Arrow / Space
- Previous Page: Page Up / Up Arrow / Left Arrow / Shift+Space
- First Page: Home / Ctrl+Home
- Last Page: End / Ctrl+End
- Go to Page: Ctrl+G

Zoom & View:
- Zoom In: Ctrl + Plus / Ctrl + Mouse Wheel Up
- Zoom Out: Ctrl + Minus / Ctrl + Mouse Wheel Down
- Reset Zoom (100%): Ctrl + 0
- Fit to Width: Ctrl + 2
- Fit to Page: Ctrl + 1
- Rotate Clockwise: Ctrl + R
- Toggle Sidebar: F4
- Fullscreen: F11

Text & Search:
- Select Text: Click and drag with mouse
- Copy Text: Ctrl + C or Right-Click -> Copy
- Find in Page: Ctrl + F
- Find Next: Enter / F3
- Find Previous: Shift+Enter / Shift+F3

Architecture & Optimization

How PrincePDF achieves instant responsiveness:

1. Asynchronous & LRU Cached Rendering:
Rendered page pixmaps are cached in memory using a Least Recently Used policy.
Only visible pages plus immediate buffer neighbors are rendered at high resolution.

2. Native C++ MuPDF Engine:
PyMuPDF provides direct bindings to the high-performance MuPDF rendering engine,
delivering 5-10x faster rasterization than browser-based PDF renderers.

3. High-DPI Aware Text Selection:
Vector character coordinates are extracted directly from PDF font glyphs.
This enables exact highlighting and clipboard text fidelity.

4. Windows 11 Native Integration:
Registered in Windows Registry (HKCU) with proper ProgID and capabilities,
allowing users to set PrincePDF as their default PDF handler in Windows Settings.